

The fibred bijection of Lenart–Sottile’s Corollary 4 exists, and its existence is empty

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Abstract

Lenart and Sottile ([math/0202090](#), Corollary 4 and the remark following it) observe that $I_\alpha(u, w) = \sum_v c_{u,v}^w I_\alpha(w_0 v, w_0)$ and write that a type-preserving map $\Gamma(u, w) \rightarrow \bigsqcup_v \Gamma(w_0 v, w_0)$ with fibre cardinality $c_{u,v}^w$ “would give a combinatorial interpretation for $c_{u,v}^w$, and thus solve the Littlewood–Richardson problem.” We prove that such a map always exists, that the number of them is an explicit multinomial product, and hence that *the existence of the map is logically equivalent to the counting identity it is derived from* and gives no interpretation of $c_{u,v}^w$ whatsoever (Theorem 5). We prove further that the fibre-cardinality condition is vacuous for $n \leq 5$, because every Schubert structure constant of Fl_n is 0 or 1 for $n \leq 5$, with $c = 2$ first occurring at $n = 6$ (Theorem 8). We record that the bijectivization-uniqueness clause of Petrov ([2609.18502](#), §5) does not apply (Proposition 10), and that the natural candidate map — the identity on second-coordinate words — is refuted. The upshot is a correction of the question: what is open is not existence but *canonicity*, and the smallest instances with any fibre content live in $\mathfrak{S}_6$.

1 The statement, and the conventions it is stated in

1.1 Source

[math/0202090](#) (Lenart–Sottile, *Skew Schubert polynomials*) is held at **deep-read**. The theorem numbers were resolved from the compiled `.aux` of `skewschub.tex` (the L^AT_EX source carries a single counter shared across theorem/proposition/lemma/corollary/example/remark, so numbers *must not* be counted by hand): `cor:identity` = {4}{4}, i.e. Corollary 4 on p. 4, and `thm:skew` = {2}{4}. The quoted remark follows Theorem 2 at l. 486–502.

1.2 Definitions

Let $\mathfrak{S}_n$ be the symmetric group, ℓ the Coxeter length, w_0 the longest element, and t_{ij} the transposition of $i < j$. Products are composition, $(pq)(i) = p(q(i))$.

Definition 1 (labelled Bruhat order; [math/0202090](#) l. 352). For each Bruhat cover $u \lessdot w$ with $u^{-1}w = t_{ij}$, $i < j$, put $j - i$ directed edges $u \xrightarrow{(k,b)} w$, one for each integer k with $i \leq k < j$, where $b = u(i) = w(j)$.

Definition 2 (increasing chain, type; l. 366, l. 372). An *increasing chain* γ from u to w is a saturated chain $u = u_0 \lessdot u_1 \lessdot \cdots \lessdot u_m = w$ together with a choice of label on each step such that the label sequence $(k_1, b_1), \dots, (k_m, b_m)$ is strictly increasing in lexicographic order on pairs. The

type of γ is the exponent vector $\alpha = \text{type}(\gamma) \in \mathbb{Z}_{\geq 0}^{n-1}$ with $\alpha_i = \#\{s : k_s = i\}$. Write $\Gamma(u, w)$ for the set of increasing chains from u to w , $\Gamma_\alpha(u, w)$ for its type- α part, and $I_\alpha(u, w) = |\Gamma_\alpha(u, w)|$. Note $|\alpha| = m = \ell(w) - \ell(u)$.

Let $c_{u,v}^w$ be the Schubert structure constants of $H^*(\text{Fl}_n)$, $\sigma_u \sigma_v = \sum_w c_{u,v}^w \sigma_w$; they vanish unless $\ell(u) + \ell(v) = \ell(w)$. The relevant duality is $c_{u,v}^{w_0} = \delta_{v, w_0 u}$ (verified below), so “ $w_0 v$ ” in Corollary 4 means $w_0 \cdot v$.

Theorem 3 (math/0202090, Corollary 4). *For all $u \leq w$ in $\mathfrak{S}_n$ and every type α with $|\alpha| = \ell(w) - \ell(u)$,*

$$I_\alpha(u, w) = \sum_{v : \ell(v) = \ell(w) - \ell(u)} c_{u,v}^w I_\alpha(w_0 v, w_0).$$

1.3 The conventions are validated, not assumed

Reading a convention out of locators is where this kind of work goes wrong, so the conventions above were *tested before use*. Specialising Theorem 2 of math/0202090 to $w = w_0$ predicts

$$\mathfrak{S}_u(x) = \sum_{\gamma \in \Gamma(u, w_0)} x^{\delta - \text{type}(\gamma)}, \quad \delta = (n-1, n-2, \dots, 1, 0), \quad (1)$$

with $\mathfrak{S}_u$ the Schubert polynomial. Computing the right side from the chain definitions above and the left side independently by divided differences ($\mathfrak{S}_{w_0} = x_1^{n-1} \cdots x_{n-1}$, $\mathfrak{S}_{ws_i} = \partial_i \mathfrak{S}_w$):

	permutations	agreeing	chains enumerated
$n = 2$	2	2	2
$n = 3$	6	6	7
$n = 4$	24	24	41
$n = 5$	120	120	393
total	152	152	443

This is a genuine cross-mechanism check: the chain side uses only the labelled Bruhat order, the polynomial side only divided differences. An earlier reading in which the monomial was $x^{\text{type}(\gamma)}$ rather than $x^{\delta - \text{type}(\gamma)}$ failed this test at once (104 of 120 permutations disagreeing at $n = 5$), which is how the correct convention was found. Everything downstream rests on the validated version.

2 Existence is trivial

Definition 4. Fix n and $u \leq w$ in $\mathfrak{S}_n$, $m = \ell(w) - \ell(u)$, and let $T = \bigsqcup_{v: \ell(v)=m} \Gamma(w_0 v, w_0)$, a disjoint union with components indexed by v . A *Corollary-4 map* for (u, w) is a function $f : \Gamma(u, w) \rightarrow T$ such that

- (i) f is *type-preserving*: $\text{type}(f(\gamma)) = \text{type}(\gamma)$; and
- (ii) for every v with $\ell(v) = m$ and every $C \in \Gamma(w_0 v, w_0)$, $|f^{-1}(C)| = c_{u,v}^w$.

This is exactly the object the remark after Theorem 2 asks for.

Theorem 5. *For every n and every $u \leq w$ in $\mathfrak{S}_n$, Corollary-4 maps for (u, w) exist, and their number is*

$$N(u, w) = \prod_{\alpha} \frac{I_{\alpha}(u, w)!}{\prod_v (c_{u,v}^w!)^{I_{\alpha}(w_0v, w_0)}},$$

the products over all types α with $|\alpha| = m$ and all v with $\ell(v) = m$. Moreover the existence of a Corollary-4 map is equivalent to Theorem 3.

Proof. A chain has exactly one type, so $\Gamma(u, w) = \bigsqcup_{\alpha} \Gamma_{\alpha}(u, w)$ and likewise the type- α part of the v -component of T is $\Gamma_{\alpha}(w_0v, w_0)$. By (i), f is therefore the disjoint union over α of maps

$$f_{\alpha} : \Gamma_{\alpha}(u, w) \longrightarrow P_{\alpha} := \bigsqcup_v \Gamma_{\alpha}(w_0v, w_0),$$

and conditions for distinct α constrain disjoint parts of the data, hence are independent. Fix α . Specifying f_{α} is the same as specifying its fibres, i.e. a family of pairwise disjoint subsets $\{B_C\}_{C \in P_{\alpha}}$ of $\Gamma_{\alpha}(u, w)$ with union $\Gamma_{\alpha}(u, w)$; condition (ii) says $|B_C| = c_{u,v}^w$ whenever $C \in \Gamma_{\alpha}(w_0v, w_0)$. (For v with $c_{u,v}^w = 0$ every such B_C is empty, so the v -component receives nothing, as it must.) Such a family of prescribed sizes exists if and only if the sizes sum to the size of the set,

$$\sum_{C \in P_{\alpha}} |B_C| = \sum_v c_{u,v}^w I_{\alpha}(w_0v, w_0) \stackrel{!}{=} I_{\alpha}(u, w),$$

which is precisely Theorem 3 for this α ; and in that case the number of such families is the multinomial coefficient $I_{\alpha}(u, w)! / \prod_{C \in P_{\alpha}} |B_C|! = I_{\alpha}(u, w)! / \prod_v (c_{u,v}^w!)^{I_{\alpha}(w_0v, w_0)}$. Types with $I_{\alpha}(u, w) = 0$ have all blocks empty and contribute a factor 1. Multiplying over α gives $N(u, w)$, and $N(u, w) \geq 1$ exactly when Theorem 3 holds for every α . $\square$

Corollary 6. *The existence of a Corollary-4 map carries no information beyond Theorem 3, and in particular yields no combinatorial interpretation of $c_{u,v}^w$.*

Proof. Immediate from Theorem 5: the two statements are equivalent. The structural reason is worth naming. In Definition 4 the numbers $c_{u,v}^w$ appear as an *input* — they are the prescribed block sizes — and a combinatorial interpretation would have to produce them as an *output*, as the cardinality of a set defined without reference to them. Theorem 5 says the fibre condition is a bookkeeping constraint on a set partition, satisfiable by fiat once the totals agree. $\square$

Remark 7. This answers the literal target “construct, or prove obstructed, a type-preserving map $\Gamma(u, w) \rightarrow \bigsqcup_v \Gamma(w_0v, w_0)$ whose fibre over the v -component has cardinality $c_{u,v}^w$ ”: it is constructed, for every n , and the construction shows the target was mis-posed. Lenart and Sottile’s sentence is not wrong — they say such a map *would give* a combinatorial interpretation, meaning a described one. The content of the 2002 problem lies entirely in the word “combinatorial”, i.e. in canonicity, and not in existence.

3 For $n \leq 5$ the fibre condition is vacuous

Theorem 8. 1. *For $2 \leq n \leq 5$, every Schubert structure constant of Fl_n satisfies $c_{u,v}^w \in \{0, 1\}$. (Exhaustive computation; two mechanisms.)*

2. *For $n = 6$, with $u = v = (1, 3, 5, 2, 4, 6)$ and $w = (2, 4, 6, 1, 3, 5)$ one has $c_{u,v}^w = 2$.*

3. Consequently the least n for which some fibre of some Corollary-4 map has cardinality ≥ 2 is $n = 6$. For $n \leq 5$ every Corollary-4 map is a type-preserving bijection $\Gamma(u, w) \rightarrow \bigsqcup_{v: c_{u,v}^w=1} \Gamma(w_0v, w_0)$, and $N(u, w) = \prod_{\alpha} I_{\alpha}(u, w)!$.

Proof. (1) was computed exhaustively over all triples $(u, v, w) \in \mathfrak{S}_n^3$ with $\ell(u) + \ell(v) = \ell(w)$, for $n = 3, 4, 5$, by expanding $\mathfrak{S}_u \mathfrak{S}_v$ and extracting the constant term of ∂_w ; the maximum value attained was 1 in each case, with 0 triples having $c \geq 2$. The computation was cross-checked against Monk’s rule $\sigma_u \sigma_{s_r} = \sum_{a \leq r < b, \ell(ut_{ab})=\ell(u)+1} \sigma_{ut_{ab}}$ on 7624 comparisons (852 nonzero) with 0 disagreements, and a negative control in which r was replaced by a different index produced 104 disagreements at $n = 4$, so the cross-check is not vacuous.

(2) u and w are the Grassmannian permutations of $\lambda = (2, 1)$ and $\lambda = (3, 2, 1)$ for $\text{Gr}(3, 6)$, of lengths 3 and 6. Computing directly in $\mathfrak{S}_6$ by divided differences gives $c_{u,u}^w = 2$. As an independent check on a different mechanism entirely, enumerating Littlewood–Richardson skew tableaux gives $c_{(2,1),(2,1)}^{(3,2,1)} = 2$, consistent with the classical expansion $s_{21}^2 = s_{42} + s_{411} + s_{33} + 2s_{321} + s_{3111} + s_{222} + s_{2211}$, which the same enumerator reproduces term for term.

(3) is immediate from (1), (2) and Theorem 5: if all $c_{u,v}^w \in \{0, 1\}$ then every nonempty fibre is a singleton, the map is injective on each type, and the multinomial collapses to $\prod_{\alpha} I_{\alpha}(u, w)!$. $\square$

Remark 9 (the plan for this session could not have worked). The computation was scoped to “ $n = 3$ and $n = 4$, exhaustively”. Theorem 8 shows that no amount of computing in $\mathfrak{S}_3$ or $\mathfrak{S}_4$ — or $\mathfrak{S}_5$ — can exhibit a single fibre of size ≥ 2 . In that range the fibre-cardinality constraint, which is the entire content of Definition 4(ii), is *identically* the statement that f is a bijection.

4 Non-vacuity accounting

Before searching, the number of *falsifiable* instances was counted, under the criterion that was specified in advance: triples (u, w, α) with $I_{\alpha}(u, w) > 1$ and at least two distinct v contributing.

	instances of Thm 3 checked	disagreements	falsifiable
$n = 2$	3	0	0
$n = 3$	23	0	0
$n = 4$	413	0	7
total	439	0	7

So Theorem 3 is verified, with 0 failures of the per-type identity and 0 failures of the coarser total-cardinality invariant $|\Gamma(u, w)| = \sum_v c_{u,v}^w |\Gamma(w_0v, w_0)|$. But $n = 3$ has *no falsifiable instance at all*: every type- α chain set there has at most one element, so a green sweep at $n = 3$ carries exactly zero information.

Worse, the pre-specified criterion is itself too weak. All seven $n = 4$ instances have the same shape: $I_{\alpha}(u, w) = 2$, with exactly two contributing v , each with $c_{u,v}^w = 1$ and $I_{\alpha}(w_0v, w_0) = 1$. Both fibres are singletons; there are $2! = 2$ Corollary-4 maps and nothing in Definition 4 to prefer either. The criterion with content is *some* $c_{u,v}^w \geq 2$, and by Theorem 8 that first occurs at $n = 6$.

Explicitly, over all pairs $u \leq w$ the number of Corollary-4 maps is: $N(u, w) = 1$ for all 23 pairs at $n = 3$; and at $n = 4$, $N = 1$ for 334 pairs and $N = 2$ for the 7 pairs above. Every constructed map was re-verified against both clauses of Definition 4 independently of its construction (23/23 and 341/341 reported OK).

5 The uniqueness clause does not apply

Proposition 10. *The bijectivization-uniqueness clause of 2609.18502 §5 — “A bijectivization always exists. . . It is unique only when one of A, B is a singleton” (l. 3497, read at source) — does not apply to Theorem 3, except on the locus $I_\alpha(u, w) = 1$ where $N(u, w) = 1$ holds trivially.*

Proof. Name the object on each side. In 2609.18502, a bijectivization of an identity $\sum_{a \in A} \text{wt}(a) = \sum_{b \in B} \text{wt}(b)$ is a coupling of the two sides: a joint measure on $A \times B$ with the prescribed marginals, presented by stochastic kernels $\mathbf{p}^{\text{fwd}}, \mathbf{p}^{\text{bwd}}$. The paper fixes the scope of A and B in its own sentence: “ A and B always mean the supports of the two sums: configurations of weight zero are discarded, and all stochasticity and uniqueness statements are imposed on the positive-weight support only.” So the hypothesis is $|A| = 1$ or $|B| = 1$ as sets of configurations.

For Theorem 3 at a fixed type α , the supports are $A = \Gamma_\alpha(u, w)$ and $B = \bigsqcup_v \{1, \dots, c_{u,v}^w\} \times \Gamma_\alpha(w_0 v, w_0)$, with all weights equal to 1. Hence $|A| = |B| = I_\alpha(u, w)$, and the hypothesis holds iff $I_\alpha(u, w) = 1$. That is the degenerate locus, and by Theorem 5 it is exactly where $N = 1$ anyway.

Two further mismatches hold even on that locus. First, the object the clause makes unique is a *coupling*, not a bijection; the paper’s own application is explicit about this — after collapsing one side to “a single term” it concludes that the pair is “drawn with probability proportional to its weight” (l. 4229–4233). A coupling is a bijection only in the zero-one extreme, which for $|A| = 1$ forces $|B| = 1$ as well. Second, the framework contains no type-preservation constraint, so even a unique coupling would not be a Corollary-4 map in the sense of Definition 4(i).

A quantitative replacement for the clause: with all weights 1 and $|A| = |B| = N$, the couplings of the two sides form $N \cdot \mathcal{B}_N$, N times the Birkhoff polytope of doubly stochastic $N \times N$ matrices, of dimension $(N - 1)^2$, whose vertices are precisely the bijections. Uniqueness holds iff $N = 1$. $\square$

Remark 11. The error the brief for this session made — and it is the reason this phase was scheduled — was to read “Corollary 4’s left-hand side is a single $I_\alpha(u, w)$ ” as the singleton hypothesis. A single *term* of an algebraic identity is not a singleton *support*: $I_\alpha(u, w)$ is the cardinality of A , not evidence that $|A| = 1$. The two readings share the word and nothing else.

6 A negative result on the natural candidate

Two structural facts make the chains unexpectedly rigid. Both were verified exhaustively for $n = 3, 4, 5$ (23, 420 and 17507 chains):

1. A chain is determined by its source together with its label sequence: 0 *collisions*.
2. Because labels increase lexicographically, the first coordinates increase weakly, so the type α determines the first-coordinate *sequence* exactly: 0 violations.

Hence a chain in $\Gamma_\alpha(u, w)$ is faithfully encoded by its second-coordinate word $b(\gamma) = (b_1, \dots, b_m) \in [n]^m$, and *both sides of Theorem 3 are sets of words of the same length in the same alphabet*. This makes one candidate canonical map available for free: the identity on words. It is refuted.

Proposition 12. $\{b(\gamma) : \gamma \in \Gamma_\alpha(u, w)\}$ and $\bigsqcup_v \{b(\gamma) : \gamma \in \Gamma_\alpha(w_0 v, w_0)\}^{\oplus c_{u,v}^w}$ are in general distinct multisets of words: they differ in 7 of 23 instances at $n = 3$ and in 280 of 413 at $n = 4$. (The smallest witness: $n = 3$, $u = \text{id}$, $w = (1, 3, 2)$, $\alpha = (0, 1)$, left word (2), right word (1).)

No relabelling of the alphabet repairs this: at $n = 3$ the failures demand $2 \mapsto 1$ and $1 \mapsto 2$ at $m = 1$, while at $m = 2$ the instance $u = \text{id}$, $w = (2, 3, 1)$, $\alpha = (1, 1)$ demands $(1, 1) \mapsto (2, 1)$, which is not induced by any map of letters.

7 Verification and controls

instrument	result	control
chain apparatus vs. divided differences, (1)	152/152 perms, 443 chains	wrong monomial fails 104/120
$c_{u,v}^w$: divided differences vs. Monk	7624 comparisons, 0 disagree	wrong r : 104 disagree
$c = 2$ at $n = 6$: $\mathfrak{S}_6$ vs. LR tableaux	both 2	LR enumerator 9/9 known values
Corollary 4	439 instances, 0 disagree	total-cardinality invariant 0 fail
constructed maps	364/364 re-verified OK	see below

The three controls planted on the search, one per claim the apparatus makes:

C1, wrong type (target type permuted). Fired on 9/23 at $n = 3$ and 175/341 at $n = 4$.

The silences are accounted for exactly, with quantifiers: 4 and 128 pairs have no chains at all (not applicable); 8 and 28 have a shift-invariant type vector (the perturbation is the identity); and 2 and 10 have $\alpha \mapsto I_\alpha(u, w)$ *constant* ($\equiv 1$) on a shift-closed support, so permuting types permutes the support and preserves every count. This last class is not a missed bug and not a vacuous control but a *vacuous instance*: on such a pair *no* type-permutation control can fire, and there are 10 of 19 such pairs at $n = 3$ and 38 of 213 at $n = 4$.

C1b, wrong type of wrong degree (non-dodgeable). Because $|\alpha| = \ell(w) - \ell(u)$ is forced, perturbing the degree cannot be absorbed. Fired on 19/19 and 213/213 applicable pairs.

C2, perturbed $c_{u,v}^w$ (one value incremented by 1). Fired on 19/19 and 213/213 applicable pairs; 0 silences.

C3, chain enumerator off its domain. $\ell(w) < \ell(u)$: 0 of 13 and 0 of 235 pairs returned a chain. $u \not\leq w$ in Bruhat order with $\ell(w) > \ell(u)$: 0 of 46 at $n = 4$ returned a chain. Positive control ($u \leq w$ must give chains): 0 empty of 19 and 213.

8 What is actually open

Theorem 5 removes existence from the problem and Theorem 8 removes $n \leq 5$ from the search space. What remains is the question Lenart and Sottile were actually pointing at, which should be stated with the quantifier in the right place.

Question 13. Is there a rule — a map defined uniformly in (u, w, α) by a combinatorial operation on chains, not referring to $c_{u,v}^w$ — which for every $u \leq w$ produces a Corollary-4 map? Equivalently: can the $N(u, w)$ candidates of Theorem 5 be cut to one by a coherence condition, so that $c_{u,v}^w$ becomes the *output* (a fibre cardinality of a described map) rather than the input?

Three things learned here constrain the search for such a rule.

1. The natural ambient set is the set of second-coordinate words: both sides of Theorem 3 embed in $[n]^m$ faithfully, and the first coordinates are determined by α and carry no freedom. A rule should be an operation on words.
2. It is not the identity on words (§6), and not induced by any relabelling of letters.
3. The first instances where the fibre condition says anything at all are in $\mathfrak{S}_6$, where $c = 2$ appears via $\text{Gr}(3, 6) \subset \text{Fl}_6$. Any candidate rule must be tested there; $\mathfrak{S}_4$ cannot distinguish a correct rule from an arbitrary one, because at $n \leq 5$ every Corollary-4 map is a bijection and $N(u, w) = \prod_{\alpha} I_{\alpha}(u, w)!$ counts arbitrary relabellings.

Gaps

1. Theorem 8(1) is a finite exhaustive verification for $n \leq 5$, not a proof for all $n \leq 5$ by argument. It is cross-checked by two mechanisms with a firing negative control, but it is a computation. (The general statement “ $c_{u,v}^w \leq 1$ for $n \leq 5$ ” presumably has a proof; I did not find one and did not look for one.)
2. Theorem 3 itself is verified here only for $n \leq 4$ (439 instances), and is quoted from `math/0202090` for general n . I did not reprove it.
3. Question 13 is untouched. Nothing in this note bears on whether a canonical rule exists; what is established is only that the existence formulation cannot be the question, and that $n \leq 5$ cannot be the test.
4. No computation was done at $n = 6$ beyond the single structure constant of Theorem 8(2). In particular the chain sets $\Gamma_{\alpha}(u, w)$ in $\mathfrak{S}_6$ for the $\text{Gr}(3, 6)$ permutations were *not* enumerated, so the first genuinely informative instance of Definition 4 has not been examined. That is the next step.

Apparatus

`proofs/code-1006c1-ls-cor4/`: `ls.py` (labelled Bruhat order, increasing chains, types), `schub.py` (Schubert polynomials by divided differences, structure constants), `lr.py` (independent Littlewood–Richardson enumerator), `cor4.py` (Corollary 4 and the non-vacuity gate), `search.py` (construction, independent re-verification, map count). Pure Python throughout, deliberately not Sage, so that the ground truth runs on a different mechanism from `sage-combinat`.